Surendra Shiwakoti

Phone number: (+49) 1782795863 (Mobile) | **Email address:** shiwakotisurendra@gmail.com | **LinkedIn:** https://www.linkedin.com/in/geosuren/ | **Address:** Stolberger Str. 4, 50933, Köln, Germany

RESEARCH INTEREST

Earth observation, Spatial data analytics, Geo-AI, Climate change adaptation, Disaster risk management

EDUCATION

MASTER OF SCIENCE IN GEOINFORMATICS 01/09/2019 – 15/09/2022

Institute for Geoinformatics, University of Münster, Germany

Thesis: *Identifying Urban Growth Patterns and Urban Heat Island Effects of Bharatpur Metropolitan City using Landsat time series imagery*

- Analyzed Landsat 8 and 9 imagery (2015–2021) to study urban expansion and Urban Heat Island (UHI) dynamics in Nepal.
- Applied remote sensing and GIS techniques including land-use/land-cover classification, Land Surface Temperature (LST) retrieval, and change detection using Python, QGIS, and Google Earth Engine.
- Quantified relationships between urban growth, vegetation loss, and surface temperature, identifying significant warming trends in urbanized zones.
- Developed geospatial workflows for image preprocessing, classification accuracy assessment, and spatiotemporal analysis of urbanization patterns.

Major Studies: Geoinformation in Society, Spatiotemporal Data Analysis, GIS, Spatial Information Infrastructure, Geosimulation, Machine Learning for Spatial Data

BACHELORS IN GEOMATICS ENGINEERING 01/05/2013 – 31/12/2017

Tribhuvan University, Institute of Engineering, Pashchimanchal Campus, Nepal

Thesis: *Land Use Land Cover Classification and Urban Density Prediction Mapping of Dolakha District*

- Processed satellite imagery, topographic maps, and spatial datasets for land-use/land-cover analysis.
- Developed urban density prediction maps using spatial indicators and GIS-based modeling techniques.
- Applied remote sensing workflows for image classification, accuracy assessment, and cartographic map production.

Major Studies: GIS, Photogrammetry, Cartography, Geodesy, Cadastral Surveying, Remote Sensing, Spatial Databases, GNSS, Digital Terrain Modeling

WORK EXPERIENCE

RESEARCH ASSOCIATE (GEOSPATIAL SCIENCE)

COLOGNE UNIVERSITY OF APPLIED SCIENCES, KÖLN, GERMANY 01/01/2024 – CURRENT

- Process and analyze large-scale remote sensing datasets using Google Earth Engine, Python, and GIS software for climate and environmental applications.
- Develop geospatial information systems and spatial data infrastructures using PostGIS, GeoServer, GeoNode, and OGC services.
- Design web-based geospatial applications, interactive dashboards, and GeoStories for environmental decision support.
- Assist development of Climate Adaptation Portal including system architecture, data integration, and user interface design.
- Conduct research on climate adaptation, land-use change, and water-resource management using geospatial analytics.
- Support stakeholder collaboration, training, and participatory geospatial workflows for municipalities and researchers.

RESEARCH SOFTWARE ENGINEER

52°NORTH SPATIAL INFORMATION RESEARCH GMBH, MÜNSTER, GERMANY 01/03/2023 – 31/12/2023

- Integrated machine learning and route optimization module into maritime Weather Routing Tool.
- Developed Python-based geospatial pipelines using GeoNode, GeoServer, PostGIS, and REST APIs.
- Created openseamap database and ingestion method for weather-aware maritime routing services.
- Conducted research on cloud-native management of Earth observation data using Open Data Cube.
- Processed large-scale geospatial datasets from OpenStreetMap and environmental data sources for spatial analytics.
- Contributed to MariData and MariGeoRoute research projects on routing optimization and geospatial intelligence.

STUDENT ASSISTANT AND RESEARCHER

52°NORTH SPATIAL INFORMATION RESEARCH GMBH, MÜNSTER, GERMANY 01/12/2021 – 30/09/2022

- Supported geospatial data management and maritime navigation research in the MariData project.
- Worked with OpenStreetMap, GeoNode, GeoServer, PostGIS, Python, and OGC web services.
- Assisted in development and testing of WebGIS and navigation applications using QGIS and Python.

STUDENT INTERNSHIP

GERMAN AEROSPACE CENTER (DLR), MÜNCHEN, GERMANY 15/03/2021 – 17/08/2021

- Developed SpatioTemporal Asset Catalogs (STAC) using pystac for Earth observation data management.
- Extended UKIS-PySAT Python tools for satellite data access and processing.
- Worked with Cloud Optimized GeoTIFF (COG) datasets for scalable remote sensing workflows.
- Used AWS S3 and boto3 for cloud-based geospatial data handling and storage.

TEACHING ASSISTANT (GEOMATICS)
KATHMANDU UNIVERSITY, DHULIKHEL, NEPAL 21/08/2018 – 17/09/2019

- Assisted teaching in surveying, cartography, engineering drawing, and geomatics fundamentals.
- Conducted field training in GNSS, surveying instruments, and geospatial data collection.
- Supervised student projects involving spatial data analysis and reporting.
- Supported syllabus development and departmental academic activities.

TRAININGS AND WORKSHOPS

ERASMUS+ TRAINING / INTERNATIONAL WORKSHOP (Climate Action Live Lab, Erasmus+, Le Créneau, France)

- Climate justice and climate anxiety
- Sustainable development frameworks
- Participatory mapping for climate risk and resilience

RESEARCH & FIELD-BASED TRAINING

- OpenStreetMap mapping, mapathons, hackathons, and thematic mapping activities (Geomatics Engineering Association of Nepal).
- Topographic surveying, drone-based data collection, and participatory mapping for environmental and social applications.
- Volunteering in environmental management and community-based geospatial data initiatives.

GEOSPATIAL, AI & PROGRAMMING (ONLINE TRAINING)

- Machine Learning for GIS and Remote Sensing (Python-based workflows)
- Google Earth Engine for land cover change detection and geospatial ML applications
- GeoAI and Deep Learning for Geospatial Analysis (CNNs for satellite image classification)
- Hyperspectral image classification using deep convolutional neural networks
- Advanced geospatial raster analysis and spatial data science in Python
- Web GIS development and deployment using GeoDjango
- ArcGIS Pro and QGIS advanced spatial analysis workflows

DATA SCIENCE & SOFTWARE ENGINEERING COURSES

- Python for Data Science and Machine Learning
- Geospatial Data Analytics (advanced Python workflows)
- Web mapping and GIS application development (end-to-end deployment)

PROJECTS AND PUBLICATIONS

FUNDED RESEARCH PROJECTS

- **CoSite –** Climate Adaptation and Resilience Strategies for Municipalities
  Funded by Federal Ministry of Education and Research
  Supported development of geospatial frameworks for climate adaptation and spatial planning.

- **MariData –** Maritime Route Optimization for Energy Efficiency
  Funded by Federal Ministry for Economic Affairs and Energy
  Contributed to route optimization constraints and geospatial-environmental modeling

- **UKIS-PySAT** – Satellite Data Processing Framework
  German Aerospace Center
  Developed and extended Python-based tools for accessing and processing satellite imagery (sentinel/ Landsat).

RESEARCH CONTRIBUTIONS

- Developed image recognition methods for citizen science applications, including movement tracking and crowd counting in built environments.
- Performed spatiotemporal analysis of satellite imagery using classification and change detection techniques in R and Python.
- Built geospatial data workflows for WebGIS and Earth observation applications using OpenStreetMap and remote sensing datasets.

PUBLICATIONS & RESEARCH OUTPUTS

- Shiwakoti, S. & Ramirez Duval, J. L. (2026). A Google Earth Engine–Based Framework for Dynamic Urban Heat Island Analysis Using Landsat 8/9 (2015–2025). Zenodo. https://doi.org/10.5281/zenodo.19921427
- Shiwakoti, S. (2022). *Identifying Urban Growth Patterns and Urban Heat Island Effects Using Landsat Time Series Imagery*. University of
  Münster. https://doi.org/10.13140/RG.2.2.23684.90248
- Hoxha et al. (2025). *Data visualization and management.* In Toolkit: Water-Sensitive Infrastructure Development. https://doi.org/10.57683/EPUB-3047

- Shiwakoti et al. (2017). *Land Use Land Cover Classification and Urban Density Prediction Mapping of Dolakha District*.

SURVEYING AND MAPPING

- Topographic, cadastral, road, bridge, and hydropower surveying
- GPS/GNSS data collection and mapping
- Differential leveling and engineering surveying
- GIS-based thematic mapping and spatial data processing
- Experience with Total Station, Level, Theodolite, Drones, and GPS instruments

SKILLS

Programming: Python, R, Matlab, JavaScript
Databases & GIS Servers: PostgreSQL, PostGIS, SQL, GeoServer
GIS & Remote Sensing: QGIS, ArcGIS, Google Earth Engine, ENVI, Pix4D
Web GIS: GeoDjango, Leaflet, OpenLayers, Mapbox, CesiumJS
Data Science & Libraries: GeoPandas, NumPy, Matplotlib, Plotly, GDAL, Rasterio
Machine Learning: Scikit-learn, TensorFlow, OpenCV
Web Development: Django, Flask, Node.js, React

PROFESSIONAL MEMBERSHIP

- Nepal Engineering Council : Professional Engineer (203 "Geomatics")
- Nepal Engineering Association: General Member
- Geomatics Engineering Association of Nepal (GESAN): General Member

LANGUAGES

- English – C1/C2 (Fluent, research and academic)
- German – B1
- Nepali – Native

ACADEMIC REFERENCES

Prof. Dr. Edzer Pebesma
edzer.pebesma@uni-muenster.de
Spatiotemporal Modelling Lab
University of Münster, Germany

Dr. Benedikt Gräler
b.graeler@52north.org
Managing Director, Research
52°North Spatial Information Research GmbH
Münster, Germany

Prof. Dr. Lars Ribbe
lars.ribbe@th-koeln.de
Cologne University of Applied Sciences
Institute for Natural Resources Technology and Management (ITT), Germany